UNIVERSITY OF LONDON

MSc/MSci EXAMINATION 2025

For Internal Students of
Royal Holloway

DO NOT TURN OVER UNTIL TOLD TO BEGIN

CS4960/CS4960R,CS5960/CS5960R: Artificial Intelligence Principles and Techniques – for FIRST SIT / RESIT CANDIDATES

Time Allowed: **TWO hours**

Please answer **ALL** questions

Calculators are not permitted

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1. (a) Given unlimited memory and time is DFS guaranteed to find a solution if there is one? Explain why. [5 marks]

   (b) Given unlimited memory and time is BFS guaranteed to find an optimal solution if there is one? Does this apply to all domains? Explain why. [5 marks]

   (c) Consider the graph in Figure 1. The initial state is 'h' and the goal state is 'c'. Draw the smallest (fewer nodes) possible BFS, DFS and IDFS search graphs and label the nodes with the order in which states are selected for expansion. For example, the first node will always be labeled as 'h1', and unexpanded nodes are only labeled with their name, e.g. 'c'. It is up to you to tie-break node expansions so that the smaller search graph is generated. [15 marks]

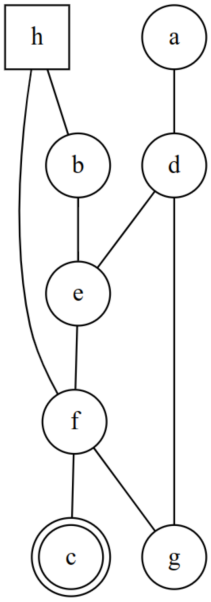


Figure 1: Connectivity Graph

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Figure 2: Central Europe Colouring Map

2. Consider the Central Europe colouring map in figure 2 where the task is to colour each region with either Red, Green or Blue so that no adjacent states have the same colour.

   (a) Formulate the map colouring problem as a *constraint network*, which consists of:

   - A set of **variables** V.
   - A set of **domains** D_v for each $v \in V$, where D_v represents the possible values that the variable v can take.
   - A set of **constraints** C_{uv} for each pair of variables $u, v \in V$, where C_{uv} specifies the allowed combinations of values for u and v.

   For the variables names use 'A' for Austria, 'C' for Czech Republic, 'G' for Germany, 'H' for Hungary, 'I' for Italy 'P' for Poland, 'S' for Slovakia, 'V' for Slovenia and 'W' for Switzerland. [10 marks]

   (b) Draw the *constraint graph* of the problem. [10 marks]

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(c) Consider the partial assignments below:

- $A = Blue$, $W = Red$, $S = Blue$;
- $P = Green$, $G = Red$, $S = Blue$;
- $C = Red$, $I = Red$, $H = Red$.

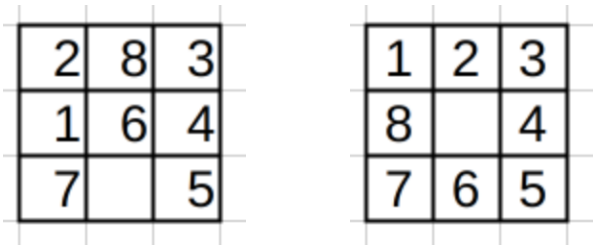
For each partial assignment, answer whether it is consistent and whether it can be extended to a solution or not; if a partial assignment is not consistent or cannot be extended to a solution, explain why. [10 marks]

(d) Describe the pre-processing technique which allows the usage of the AcyclicCG algorithm on a graph with cycles. Make sure to explain what is required for the technique to be complete, i.e. if there is a solution it will be found eventually. [5 marks]

(e) Modify this coloring problem so that now there are four possible colours per country (Red, Green, Blue and Yellow). Use forward propagation for the following partial assignment: $A = Red, C = Green, W = Green$ on the remaining variables and write down the corresponding domains for the remaining variables. Then, draw the resulting constraint graph without Austria, Czech Republic and Switzerland. Finally, use AcyclicCG to find a satisficing solution for the remaining variables. Remember to both speficy the variable ordering and draw the corresponding tree. Pick G as the root variable for the tree. [20 marks]

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3. Consider the following Eight-puzzle problem, where Figure 3a shows the initial state and figure 3b shows the goal state. The empty(blank) tile can exchange its position with any vertically or horizontally adjacent tile, e.g. for the initial state only tiles “7”, “6” and “5” can exchange their position with the empty tile.



(a) Initial State (b) Goal State

Figure 3: Initial and goal states for the Eight-Puzzle

Using as a heuristic the number of misplaced tiles, draw the A* search graph. For each node, draw or write the corresponding tile placement. Each node must also be labeled with its h, g and f values. You do not need to write down the expansion order and you can tie-break according to your preference. Stop the A* search once a goal state is generated.

[20 marks]

END

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